Toward a four-dimensional Yang–Mills construction

Notebook 29: total Ward extraction and classical spatial gain

Research working document

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Abstract

We continue the existence construction by treating the low field degrees left open by the high-degree estimate of Notebook 28. The actual local macrostep preserves output translations and the hypercubic group, under the selected averaging map's stated permutation axiom. Its total correction obeys the nonlinear gauge identities. Uniform kernel locality permits a joint subsequential limit of vacuum-normalized analytic interactions and coherent classical background families. There, the quadratic, cubic and quartic marginal vertices are all fixed by one coefficient multiplying a normalized gauge-invariant reference action. The actual homogeneous classical action has the required normalization, so this extraction has an exact algebraic coupling rebase.

We also propagate complete first-gradient Hölder regularity through the ordinary averaging prefix, including strict support families. Using total moment cancellations before assigning carriers, an exact lattice factorization puts one difference on each quadratic factor. Together with cubic and quartic spatial subtraction, this gives a connected-activity bound of order $C(J)L^{-\alpha}$ for the canceled low-degree background, under the specified coherent-family and infinite-kernel hypotheses. The coefficient $C(J)$ is retained: it cannot be defeated by a ratio chosen before J. Two ordinary minimum comparisons prove a classical background semigroup modulo gauge. This allows a finite endpoint ratio $L = R^q$ to be chosen after fixing the elementary ratio R and grouping J. An activity space modulo equality of normalized totals is Banach. The full-reference extraction is bounded independently of the small analytic radius, and classical background substitution descends to this quotient. Combining the low- and high-degree estimates gives a classical contraction under the stated joint map and support hypotheses.

These results give a gauge-compatible marginal classification, a quantitative classical localization of its remainder, and a complete space in which the representation can change. They do not yet establish that the actual quantum update descends to this space with the required finite-regulator compatibility, the required coupling trajectory, or the remaining large-field and continuum construction.

The objective remains a complete existence proof for a nontrivial four-dimensional Yang–Mills theory. The preceding notebooks establish the homogeneous classical branch and a local quantum macrostep. This notebook distinguishes exact finite-regulator local identities from jointly extracted infinite analytic interactions and from the operator estimates that require those coherent limiting families. The compactness argument connects these levels at fixed parameters; it does not provide a universal quantum update on all such limits. Gauge invariance is required of the total functional, not of each chosen activity coefficient.

1 Actual macrostep symmetries and the quadratic tensor

This section concerns the actual local composite sector and its common-domain analytic continuation. It does not assert equality of different gauge sectors or spatial symmetry of an arbitrary measurable completion outside the analytic domain. The input total density must have the spatial symmetries under consideration.

1.1 Symmetries of the exact local integral

Fix the odd primitive ratio ℓ, a finite power $R = \ell^s$, and the centered nested blocks of the composite construction. Use compatible *cubic* tori, with distinct lifted interfaces as in the geometric proof. The sufficiently large auxiliary grouping J can be chosen as a power of ℓ, preserving a cofinal family of source tori. On rectangular tori only permutations preserving the side lengths are available.

The selected source mean is assumed to obey its permutation axiom as well as the endpoint covariance already used in the construction. The all-coordinate-order contour family and this axiom are the properties stated in Bałaban, CMP 109 (1987), pp. 252–254, especially (0.7). The implicit mean (0.10) is an admissible example; it is not the definition of every permitted mean.

Proposition 1.1 (Exact spatial symmetry of the local sector)**.** *Suppose the complete input density is invariant under the relevant spatial transformations on the common small domain. The composite constraint, independent direct-R internal contour gauge, physical centered branch and actual retained link/color-ball integral commute with*

$$x \longmapsto Sx + Rn, \qquad n \in \mathbb{Z}^d, \quad S \text{ a signed coordinate permutation.} \tag{1}$$

After the output lattice is rescaled by R, these give all its unit translations and the full hypercubic point group. The same symmetries hold for the total generated correction on the common zero-free holomorphic domain, with its compatible logarithm branch.

Proof. Every transformation in (1) preserves each level of the centered hierarchy. It permutes the coordinate-order contour family. The mean's permutation axiom therefore makes the contour maps equivariant. Orientation reversal of an ordinary coarse bond also respects the averaging formula. If that formula is written as

$$F = \exp\left\{ N^{-1} \sum_x \log(U_x T^{-1}) \right\} T,$$

the reversed strings and reference are U_x^{-1}, T^{-1}, and

$$\log(U_x^{-1} T) = -\operatorname{Ad}_{T^{-1}} \log(U_x T^{-1}), \qquad F_{\text{reverse}} = F^{-1}. \tag{2}$$

The primitive maps, their composition and the separate direct-R contour map consequently commute with the transformation. The centerline interface pivots are preserved as unoriented bonds.

The relative-coordinate calculation is essential. For $U_b = e^{iB_b} V_b$, a reversed bond satisfies

$$U_b^{-1} = e^{-i \operatorname{Ad}_{V_b^{-1}} B_b} V_b^{-1}. \tag{3}$$

At fixed real V, this is an orthogonal linear color map. Together with the bond permutation it induces an orthogonal map $D_{S,V}$ on the retained coordinates. Uniqueness of the small pivot solution gives

$$\Phi_{SV}(D_{S,V} z) = \mathcal{R}_{S,V} \Phi_V(z), \tag{4}$$

where $\mathcal{R}_{S,V}$ is the full relative-bond transformation. Thus the original window $\prod_{b \text{ retained}} 1_{\{|z_b| \le gT\}}$ is preserved with absolute Jacobian one. No complex change of a sharp indicator is used.

The transformed physical branch is feasible, has all internal contours equal to the identity, and has the same constrained classical value and stationarity. Common-domain uniqueness yields $V_*(SW) = SV_*(W)$. The homogeneous classical action has the required value symmetry by applying the same transformations to its defining ordinary-constraint minimization; the fine Wilson action and small global norm class are preserved.

Haar factors are invariant under inverses and conjugation. A pivot derivative transforms by the relative coarse and pivot color maps. On a reversed bond both have determinant $(-1)^{\dim \mathfrak{g}}$; their

adjoint factors have determinant one. The signs cancel in the ratio. Permutations merely relabel the factors. This accounts for both the background delta determinant and the normalized straightening determinant.

The chart action plus absolute gauge penalty satisfies $T_{SV}(D_{S,V}z) = T_V(z)$. The map $D_{S,V}$ is independent of z, so constant, linear and quadratic Taylor terms transform separately, also in the centered independent-background extension. The Hessian transforms by orthogonal congruence, preserving the normalized Gaussian and its determinant. The fixed flat determinant is the same scalar denominator. The retained change of variables therefore proves equality of the real local integrals.

The complete inherited insertion agrees with its pre-gauge representation on the original common domain. Its assumed spatial invariance and holomorphic uniqueness extend the total identity to the common-domain continuation. No symmetry of its individual activity coefficients is required. Finally the proved zero-free cluster domain and common branch give equality of logarithms. Holomorphy alone would not imply this last step. □

Auxiliary J-grouping reorganizes the exact retained product window and finite integral; it does not alter their scalar total. Its fixed carrier grid and deterministic trees may break symmetry at coefficient level. The proposition concerns their sum. Additional genuine block-norm or asymmetric cutoff restrictions would require a new check. A color-coordinate cube is not a substitute for the invariant linkwise color balls.

There is an explicit obstruction to current-lattice unit translations for the fixed sector. A pure gauge with a small nonzero phase t at one block root and zero elsewhere has absolute contour penalty

$$(R^d - 1)q(e^{it}), \qquad q(U) = 1 - \tfrac{1}{2}\tau(U + U^{-1}).$$

Moving that phase one current unit to a nonroot gives only $q(e^{it})$. This does not conflict with output unit translations, whose current lift is R.

The measurable completion using an ordered reference contour and fallback is gauge equivariant, but its global spatial equivariance has not been proved. The proposition does not concern that completion or the uncontrolled complement. If needed, averaging transformed sector–complement decompositions preserves their exact sum when the blocking map and input are equivariant. Averaging conjugated asymmetric blocking kernels instead defines a new Markov blocking kernel; it is not equality with the original deterministic map. No such change is made here.

Charge conjugation. For $SU(2)$, $U \mapsto \overline{U}$ is conjugation by $i\sigma_2$, hence is already a constant gauge transformation. It is exact on the same domains. For $SU(N)$, the holomorphic automorphism is $U \mapsto (U^{-1})^T$, with orthogonal real Lie map $A \mapsto -A^T$. The implicit mean $\sum_j \log(U_j P^{-1}) = 0$ commutes with this automorphism by local uniqueness. The proof then applies if the complete input is also invariant. For a different unspecified allowed mean, this outer-automorphism covariance is an extra property to verify; reality alone is not a proof.

1.2 Uniform Hessian decay and a kernel limit

Let the *total* output correction have connected activities on output J-cells:

$$F(w) = \sum_{H \text{ connected}} F_H(w_H), \qquad \sup_I \sum_{H \ni I} e^{a_0|H|} \|F_H\|_b \le \epsilon. \tag{5}$$

Here the supremum is on a scalar-coordinate polydisc of radius b; a color-ball domain may be replaced by a contained scalar polydisc. Write $D_0 = d(\dim \mathfrak{g})J^d$ for an upper bound on the scalar coordinate count in a cell. All parameters are uniform in volume and preceding homogeneous depth; J is fixed.

For bond/color coordinates i, j, let ξ_i, ξ_j be their bond midpoints and $K_{ij} = \partial_i \partial_j F(0)$. Cauchy gives

$$|\partial_i \partial_j F_H(0)| \le 2b^{-2} \|F_H\|_b.$$

If connected H contains both cells, their midpoint distance is at most $J(|H| + d)$, using shortest distance on a torus. Sum the at most $D_0|H|$ second coordinates before the anchored support sum. With $\mu_0 = a_0/(2J)$, this proves

$$\sup_i \sum_j e^{\mu_0 |\xi_i - \xi_j|_1} |K_{ij}| \le \mathcal{M} := \frac{2D_0 e^{a_0 d/2} \epsilon}{b^2} \sup_{n \ge 1} n e^{-a_0 n/2}. \tag{6}$$

No symmetry of the activity decomposition was used.

For the real quadratic classification assume F is real on real small fields, as for the physical positive density and its real logarithm. Complex inputs have the same algebraic classification with a possibly complex β. For a compact semisimple group, constant gauge invariance gives $DF(0) = 0$. Differentiate local gauge invariance once in a site gauge and once in a field direction. The derivative of its nonlinear generator is multiplied by $DF(0)$, so the exact position-space Ward identity is

$$KD = 0, \tag{7}$$

with D the output unit gradient.

Take an exhausting sequence of compatible cubic tori and assume the uniform bound (6). A diagonal subsequence gives a kernel limit at every fixed displacement. Fatou preserves exponential summability. Translation covariance, signed permutations, Hessian symmetry and the finite-difference identity (7) pass at fixed displacements. This supplies an infinite-lattice kernel, not a probability measure, uniqueness statement or continuum limit.

This limiting step is substantive. Finite periodic Ward identities hold only at their allowed momenta and do not by themselves imply a vanishing zero-momentum mass coefficient. Small gauge-invariant winding terms are possible. No interpolation of finite discrete momenta is used below.

1.3 Midpoint Fourier classification with a remainder bound

Constant adjoint invariance makes the color structure scalar on each simple ideal. Cross terms between distinct simple ideals vanish. Work with one such spatial kernel $k_{\mu\nu}(r)$. Define

$$\widetilde{K}_{\mu\nu}(p) = \sum_r k_{\mu\nu}(r) e^{ip\cdot(r + (e_\nu - e_\mu)/2)}, \qquad \widehat{p}_\mu = 2\sin(p_\mu/2). \tag{8}$$

The exponential bound makes the symbol analytic near zero. The limiting position-space Ward identity yields

$$\widetilde{K}(p)\widehat{p} = 0 \quad \text{for every real } p. \tag{9}$$

At $p = te_\mu \ne 0$, this kills e_μ; continuity gives $\widetilde{K}(0) = 0$. Full inversion gives $\widetilde{K}(-p) = \widetilde{K}(p)$. Reality and Hessian symmetry make the symbol real symmetric for real p. Midpoint coordinates give the phase-free signed-permutation law and eliminate odd Taylor degrees.

Theorem 1.2 (Quadratic Maxwell tensor)**.** *Under the preceding total symmetries, Ward identity and uniform kernel hypotheses, each simple color factor has a real coefficient β such that*

$$\widetilde{K}(p) = \beta(|p|^2 I - pp^T) + O(|p|^4) = \beta(|\widehat{p}|^2 I - \widehat{p}\widehat{p}^T) + O(|p|^4). \tag{10}$$

The estimate concerns the subsequential infinite kernel; it is not an assertion about a continuous interpolation of a finite-volume symbol.

Proof. Signed reflections force the degree-two part to have only squares on its diagonal and $p_\mu p_\nu$ off diagonal. Permutations identify their coefficients:

$$(K_2)_{\mu\mu} = Ap_\mu^2 + B\sum_{\nu\neq\mu} p_\nu^2, \qquad (K_2)_{\mu\nu} = Cp_\mu p_\nu \quad (\mu \neq \nu).$$

The cubic part of (9) is $K_2(p)p = 0$. Its polynomial coefficients give $A = 0$ and $C = -B$. Put $\beta = B$. Analyticity and parity give the fourth-order remainder, and $\widehat{p} = p + O(|p|^3)$ gives its second form. □

For explicit constants set

$$M_n = \max_\mu \sum_{\nu,r} |r + (e_\nu - e_\mu)/2|_1^n \, |k_{\mu\nu}(r)|. \tag{11}$$

For $n > 0$, (6) gives

$$M_n \leq \mathcal{M}\left(\frac{n}{e\mu_0}\right)^n, \qquad |\beta| \leq M_2/2.$$

The even cosine Taylor remainder yields, for real p,

$$\|\widetilde{K}(p) - K_2(p)\|_{\mathrm{op}} \leq M_4\|p\|_\infty^4/24.$$

Using $|2\sin(t/2) - t| \leq |t|^3/24$ in every row, for $\|p\|_\infty \leq 1$ we obtain

$$\left\|\widetilde{K}(p) - \beta(|\widehat{p}|^2 I - \widehat{p}\widehat{p}^T)\right\|_{\mathrm{op}} \leq \left(\frac{M_4}{24} + \frac{d}{6}|\beta|\right)\|p\|_\infty^4. \tag{12}$$

Real symmetry turns the absolute row estimate into the operator estimate. Without coordinate permutations there need not be a single β: unequal plaquette-orientation coefficients already give a local gauge-invariant action with translations and axis reflections but several quadratic coefficients.

1.4 Scope for a subsequent extraction

The preceding argument establishes actual output-lattice symmetry of the local $SU(2)$ integral and the resulting quadratic classification. It determines neither the sign of β nor asymptotic freedom. The uniform flat classical normalization is a separate, already available input: the explicit alias formula in Notebook07 gives

$$q_{M,\mathrm{mid}}(p) = |p|^2 I - pp^T + O(|p|^4)$$

uniformly in M. Together with the homogeneous minimizer's identification of q_M as $D^2 S_M(0)$, this supplies the classical quadratic normalization in that scope. Ellipticity alone would not have supplied it.

Subtracting β times the normalized standard Wilson plaquette functional is a gauge-invariant scalar subtraction whose residual quadratic kernel satisfies (12). A localized nonlinear extraction, the higher-field Ward completion, finite-volume winding control and stable relevant/marginal flow remain additional steps. Coefficientwise high-field-jet contraction by itself does not supply them.

2 Hölder regularity through an ordinary prefix

Fix the primitive odd ratio ℓ and $0 < \alpha < 1$. Write $N = MR = \ell^{k+s}$, $\eta = N^{-1}$ and $h = \ell^k\eta = R^{-1}$. Physical next-coarse spacing is one. Define the physical prefix map

$$P_h(A) = \frac{1}{ih}\log M_\ell^k(e^{i\eta A}). \tag{13}$$

The established ordinary-trajectory theorem gives a common complex supremum ball on which P_h is holomorphic, vanishes at zero and $\|P_h(A)\|_\infty \le K\|A\|_\infty$, uniformly in M, R and compatible periodic volume. Its dependency radius is at most $c_\ell h$ in physical coordinates. Nested centered partitions give

$$P_h(T_z A) = T_z P_h(A), \qquad z \in h\mathbb{Z}^d. \tag{14}$$

Only these aligned translations are asserted. Use the complete norm

$$\|A\|_{\mathcal{N}_{\alpha,\eta}} = \|A\|_\infty + \|\nabla_\eta A\|_\infty + [\nabla_\eta A]_\alpha,$$

with physical distances and equivalent fixed-collar versions on periodic lifts. Constants below depend on the primitive ratio, α and the group, not on M, R or auxiliary grouping J.

Lemma 2.1 (Uniform complete-field prefix bound)**.** *On a smaller fixed ball,*

$$\|P_h(A)\|_{\mathcal{N}_{\alpha,h}} \le C(\|A\|_{\mathcal{N}_{\alpha,\eta}} + \|A\|^2_{\mathcal{N}_{\alpha,\eta}}). \tag{15}$$

The map is holomorphic into the complete output space with uniform Banach Cauchy derivative bounds.

Proof. Uniform supremum holomorphy bounds DP_h, D^2P_h on a smaller supremum ball by constants K_1, K_2. For $T_h = T_{he_\mu}$, $\nabla_h A = (T_h A - A)/h$ is an average of fine gradients; its supremum and Hölder seminorm are bounded by those of $\nabla_\eta A$. Equation (14) gives

$$\nabla_h P_h(A) = \int_0^1 DP_h(A + t(T_h A - A))\nabla_h A \, dt. \tag{16}$$

If $\|A\|_{\mathcal{N}_{\alpha,\eta}} \le b$, this bounds the output gradient by $K_1 b$. Translate by $z \in h\mathbb{Z}^d$ and subtract. The change in direction costs $CK_1 b|z|^\alpha$. The derivative operator changes by at most $K_2\|T_z((1-t)A + tT_hA) - ((1-t)A + tT_hA)\|_\infty$. After multiplication by $\|\nabla_h A\|_\infty$, this costs $CK_2 b^2|z|$, hence $CK_2 b^2|z|^\alpha$ when $|z| \le 1$. For larger shifts the gradient supremum suffices. This proves (15), including periodic lifts and short tori with incidences counted with multiplicity. All interpolation segments lie in the common complex ball. At each finite regulator the map is holomorphic between finite-dimensional normed spaces; the displayed uniform bound on the complete-field ball and Banach Cauchy give uniform complete-output derivative bounds. No ordinary-average physical adjoint is used. □

2.1 Supported pre-gauge fields and their differences

Let the established ordinary representative be $A_{\mathrm{o},N}(w) = T_N(A_{\mathrm{c},N}(w))$, with $\|A_{\mathrm{o},N}(w)\|_{\mathcal{N}_{\alpha,\eta}} \le C_{\mathrm{o}}\|w\|_\infty$. Define

$$a_p(w) = hP_h(A_{\mathrm{o},N}(w)), \qquad e^{ia_p} = M_\ell^k(e^{i\eta A_{\mathrm{o},N}(w)}). \tag{17}$$

This is the selected representative before the rooted current gauge. It is not a regularity assertion for every gauge-equivalent representative. Lemma 2.1 implies

$$\|a_p(w)\|_{\mathcal{N}_{\alpha,h}} \le Ch\|w\|_\infty. \tag{18}$$

In current-unit coordinates, let Δ_μ be the unscaled difference between adjacent h-lattice sites. Thus, on $\|w\| < b$,

$$\|a_p\|_\infty \le Cbh, \qquad \|\Delta_\mu a_p\|_\infty \le Cbh^2, \qquad \|\Delta_\nu\Delta_\mu a_p\|_\infty \le Cbh^{2+\alpha}. \tag{19}$$

More generally, for an integer current-lattice shift z,

$$|\Delta_\mu a_p(x+z) - \Delta_\mu a_p(x)| \le Cbh^{2+\alpha}|z|^\alpha. \tag{20}$$

For physical shifts longer than one, the gradient supremum proves the same bound. Periodic distances use compatible lifts. The second-difference estimate uses first-gradient Hölder regularity; it does not assert a uniform classical C^2 bound.

Let support labels refer to final physical unit cells and let τ be graph Steiner length. Fix an admissible spatial exponent $\kappa > 0$ from the classical support theorem and any prescribed finite cardinality reserve $a \geq 0$. For an output root i, use one complete h-field fiber $E_{h,i}$ with the local $\mathcal{N}_{\alpha,h}$ norm on a fixed enlarged collar of i, including all required difference endpoints and Hölder quotients.

Proposition 2.2 (Complete-output strict supports)**.** *There are $r_a, C_a > 0$, independent of M, R and volume, and strict holomorphic families*

$$a_p|_i = \sum_{X \neq \varnothing} a_{p;i,X}(w_X), \qquad a_{p;i,X}(0) = 0, \tag{21}$$

such that, for $0 < r \leq r_a$,

$$\sup_i \sum_X e^{\kappa\tau(\{i\}\cup X)+a|X|} \|a_{p;i,X}\|_{E_{h,i};r} \leq C_a h r. \tag{22}$$

The linear supported part has bound $C_a h r$ and the nonlinear remainder has bound $C_a h r^2$, after a fixed radius reduction.

Proof. Use the classical support theorem's smooth partition and input Banach spaces of complete supported fine fields. Reconstruction has a uniform norm C_{rec}; no sharply restricted Hölder field is extended. The map $P_h \circ T_N$ is uniformly holomorphic into $E_{h,i}$ by Lemma 2.1 and the ordinary-transfer theorem. Its row reads at most a fixed number N_0 of input physical cells in a fixed radius r_0. This follows from the $O(h)$ prefix range and the fixed physical range of T_N.

On a common complete-field radius ρ_1, Cauchy and polarization, followed by summing only the input cell slots, give the mixed degree-n majorant

$$v_n \leq K \left(\frac{e C_{\mathrm{rec}} N_0 e^{\kappa r_0}}{\rho_1} \right)^n = K c^n, \qquad n \geq 1. \tag{23}$$

The output collar can contain $O(R^d)$ bonds, but its complete norm is one Banach output norm, taken before the support sum. No output coordinate count occurs in (23).

Substitute the corrected minimizer's complete-field support family, whose norm with the prescribed reserve is at most $C_a' r$. Rooted Steiner trees glue at the input slot roots, and $e^{a|\cup_q X_q|} \leq \prod_q e^{a|X_q|}$. Hence the resulting series is bounded by $\sum_{n\geq 1} K(cC_a' r)^n$, at most $2KcC_a' r$ when $cC_a' r \leq 1/2$. Multiplication by h proves (22). Use one global Taylor/partition assignment and group by the union of input variable supports before restricting complete output coefficients to collars. Their restrictions on overlapping collars agree; arbitrary separate decompositions at different output roots are not used. Absolute convergence in the complete local output norm identifies the evaluated series with (17). Separating its linear terms gives the quadratic remainder estimate. Applying the bounded difference maps on the enlarged collars yields the weighted family versions of (19)–(20). □

These are finite-regulator coefficients and can retain dependence on the fixed regulator and baseline. Volume consistency is not asserted. A larger prescribed reserve may shrink r_a.

2.2 An oriented discrete Taylor identity

For a current-unit scalar component set $D_\nu f(v) = f(v + e_\nu) - f(v)$. Choose a fixed shortest coordinate path Γ from x to y. A traversed edge has positive direction $\nu(e)$, lower positive-coordinate endpoint

$v(e)$, and traversal sign $s(e)$. Telescoping gives

$$f(y) - f(x) = \sum_{\nu} (y_\nu - x_\nu) D_\nu f(x) + \mathrm{Rem}_{xy}(f),$$
$$\mathrm{Rem}_{xy}(f) = \sum_{e \in \Gamma} s(e) [D_{\nu(e)} f(v(e)) - D_{\nu(e)} f(x)]. \tag{24}$$

A negative traversal uses the positive difference at its lower endpoint, not at its traversal initial point. If $n = |x - y|_1$, (19) and (20) imply

$$|a_p(y) - a_p(x)| \le C b h^2 n, \qquad |\mathrm{Rem}_{xy}(a_p)| \le C b h^{2+\alpha} n^{1+\alpha}. \tag{25}$$

One also has the elementary bound $|\mathrm{Rem}_{xy}(f)| \le n^2 \sup |D^2 f|$. For a winding periodic path use its lifted displacement; no global affine function on a torus is assumed. Coherent complete coefficients justify telescoping across collars. If a subsequent estimate keeps a fixed starting-root spatial weight along a long path, the exponential comparison of root weights along that path must also be budgeted; it is not omitted by the local norm bound.

These independently derived conventions supply the scale pattern used in CMP109, pp. 285–288, (4.17)–(4.18) and (4.30). They do not assert the subsequent nonlinear Ward cancellation or its local carrier assignment.

Constant order and scope. The constants above are fixed at the primitive ratio before R and J. Quantum coordinate-kernel estimates obtained from activities on J-cells can instead contain factors depending on their dimension and on conversion of cell distance to current distance. Finiteness of those constants does not make them independent of J. When J is chosen after R, an expression $C(J)R^{-\alpha}$ is therefore not yet a noncircular contraction estimate. This regularity theorem does not classify the low-degree kernels, supply a bounded gauge-compatible extraction, determine a beta-function sign, or close the quantum induction.

3 Compactness of analytic interactions and coherent backgrounds

This is compactness of analytic local interactions and classical maps at fixed parameters. It is not compactness of probability measures or a continuum construction. It supplies a joint infinite-volume interface for the total-kernel Ward and spatial transport arguments.

3.1 A limiting interaction at the original radius

Fix a cell size J, radius $b > 0$, exponent $a > 0$, and finitely many scalar coordinates per cell. Let compatible cubic tori exhaust the lattice. Suppose their connected analytic coefficients satisfy

$$\sup_I \sum_{H \ni I} e^{a|H|} \|E_{N,H}\|_b \le \epsilon. \tag{26}$$

No consistency between volumes, or symmetry of individual coefficients, is assumed. For compactly supported fields define the vacuum-normalized total

$$\Phi_N(A) = \sum_H \big[E_{N,H}(A|_H) - E_{N,H}(0)\big]. \tag{27}$$

If K contains the cells meeting the field support, then $|\Phi_N(A)| \le 2|K|\epsilon$ for $\|A\|_\infty < b$.

Proposition 3.1 (Interaction compactness without radius loss)**.** *There is a subsequence and a connected analytic family E_H on the original open radius-b polydiscs with*

$$\sup_I \sum_{H \ni I} e^{a|H|} \|E_H\|_b \le \epsilon. \tag{28}$$

Its vacuum-normalized total Φ is absolutely convergent on compact fields. On every fixed finite field support and every radius $r < b$, Φ_N converges locally uniformly to Φ. In particular all fixed derivatives of the totals converge.

Proof. Each fixed finite connected H has an unambiguous central embedding on sufficiently large tori. Its coefficients are bounded holomorphic functions on one finite-dimensional polydisc. Montel, or diagonal extraction of bounded Taylor coefficients followed by Cauchy estimates, gives a locally uniform subsequential limit. There are countably many carriers, translates and scalar Taylor indices, so one diagonal subsequence works for all of them.

For a finite selection of carriers, Fatou at radius $r < b$ passes their weighted bound. Increase r to b, then exhaust the carriers, to obtain (28). The norm is the supremum on the open b polydisc. This retains its original radius without asserting uniform convergence at the boundary.

For carriers meeting K, the omitted large-carrier terms obey

$$\sum_{\substack{H \cap K \neq \varnothing \\ |H| > Q}} |E_{N,H}(A|_H) - E_{N,H}(0)| \le 2|K| \epsilon e^{-aQ}. \tag{29}$$

Only finitely many connected carriers of size at most Q meet K. Once the torus side is much larger than JQ and the radius of K, each has a unique central lift; winding carriers are covered by the tail. Convergence on the finite collection followed by (29) proves convergence of the totals. Finite-dimensional Cauchy then proves convergence of their fixed derivatives. □

The infinite kernels can therefore be derivatives of one analytic interaction, rather than unrelated subsequential tensors. Translations, signed permutations and reality of the totals pass by applying the convergence to a compact field and its fixed transform. For gauge invariance choose a compactly supported small site gauge v such that A, A^v lie in a smaller polydisc. The transformed field has support in the original support plus a finite edge neighborhood. Passing the identity $\Phi_N(A^v) = \Phi_N(A)$ gives the limiting identity and its differentiated Ward equations. Constant conjugation preserves compact field support and passes as well. These are identities of totals; no symmetry of E_H is asserted.

3.2 Joint compactness of coherent complete-field families

Fix a preceding homogeneous depth M and endpoint ratio L. The current output spacing in final units is $h = L^{-1}$. Suppose the finite-regulator classical maps have the coherent families of the supported pre-gauge construction:

$$a_N|_i = \sum_{X \neq \varnothing} a_{N;i,X}(w_X), \qquad a_{N;i,X}(0) = 0,$$

$$\sup_i \sum_X e^{\kappa\tau(\{i\} \cup X) + A|X|} \|a_{N;i,X}\|_{C^{1,\alpha};b} \le Cb/L. \tag{30}$$

The complete physical collar norm, including its maxima and first-gradient Hölder quotients, is inside the support sum. Coefficients with the same X restrict one global assignment and agree on overlapping collars.

At fixed L, each bounded collar has finitely many current coordinates. Apply Montel and a countable diagonal extraction to all fixed i, X, output coordinates and Taylor indices. Overlap identities pass to

the limits. Each complete collar norm is a finite collection of components and difference quotients with fixed physical denominators. It is lower semicontinuous under the convergence. Fatou thus preserves (30) with the same constants.

The sum of terms with either $\tau(\{i\} \cup X) > Q$ or $|X| > Q$ is bounded by

$$C(b/L)(e^{-\kappa Q} + e^{-AQ}).$$

The remaining supports form a finite set, since they lie in a fixed ball about i. Hence the full complete fields converge locally uniformly at each finite set of roots. The limiting support series is normally convergent and defines an analytic map a. Uniform Cauchy remainders also give its Banach-valued analyticity.

For compact final support K, any nonzero coefficient has $X \cap K \neq \varnothing$. Its rooted tree therefore joins i to K, yielding

$$\|a(w)\|_{\text{complete collar } i} \leq C(b/L)e^{-\kappa \operatorname{dist}(i,K)}. \tag{31}$$

A bound summed over cells of K differs only by that finite factor. The same estimate holds uniformly on centered finite tori with periodic distance. It gives summable output tails at fixed L, including its fixed number of current coordinates per unit cell.

This construction may be made jointly with the interaction extraction and with any countable collection of fixed M, L. A cofinal sequence of tori can be chosen divisible by the first finitely many required powers at each diagonal stage. The result supplies coherent infinite families; it does not choose a unique background independently of the subsequence.

3.3 Summable fields and actual scalar composition

The limiting interaction extends to summable coordinate fields with $\|A\|_\infty \leq \rho < b$. Coordinate Cauchy and the segment tA give

$$|E_H(A|_H) - E_H(0)| \leq \frac{C\|E_H\|_b}{b-\rho} \sum_{j \in H} |A_j|.$$

Summing first over carriers containing each coordinate yields

$$|\Phi(A)| \leq \frac{C\epsilon}{b-\rho}\|A\|_{\ell^1}, \qquad |\Phi(A) - \Phi(B)| \leq \frac{C\epsilon}{b-\rho}\|A - B\|_{\ell^1} \tag{32}$$

when both fields lie in that interior ball. Coordinate counts are already in the ℓ^1 sum; there is no volume factor. The same estimates hold on the finite regulators.

Choose L so the classical amplitude bound places $a_N(w)$ in a fixed interior radius $\rho < b$ on the final data domain. For compact w, (31) gives uniform ℓ^1 tails of $a_N(w)$. On centered representatives, the periodic distance from an output root to K is at least its centered distance from the origin minus the fixed radius of K, so this tail bound is uniform in volume. Truncate the output to a large fixed region. On the region, both the interaction coefficients and the classical fields converge locally uniformly. Outside it use (32). It follows that

$$\Phi_N(a_N(w)) \longrightarrow \Phi(a(w)) \tag{33}$$

locally uniformly on compact final supports and smaller data radii. This is convergence of actual scalar compositions, not only of formal Taylor series.

For decaying noncompact intermediate inputs, apply coordinate Cauchy directly to (30). Only supports containing the input coordinate j contribute, and their rooted tree spans i, j. On an interior ball,

$$\|\partial_{w_j} a_i\| \leq \frac{C}{b-\rho} e^{-\kappa \operatorname{dist}(i,j)}. \tag{34}$$

The fixed input and output fiber dimensions, including L^4, are absorbed in C. Summing over output roots proves an ℓ^1-to-ℓ^1 derivative bound, uniform in volume at fixed L. Segment integration gives the corresponding Lipschitz estimate. Since $a(0) = 0$, summable inputs have summable outputs.

The same truncation proof therefore applies to finite compositions, provided every intermediate field remains in a strict common interior of its analytic domain. Finite-volume scalar identities, including a classical semigroup identity, can pass to a jointly extracted finite list of maps and interactions on compatible nested volumes. Separate arbitrary subsequences need not compose.

3.4 Scope

The result jointly attains a subsequential analytic interaction and coherent classical families at fixed parameters. It supplies the total-kernel and complete-field hypotheses used by the spatial remainder theorem. It proves neither uniqueness, convergence of Gibbs measures, a continuum theory, nor consistency of a quantum update for all input families. Descent of the actual quantum map to a chosen interaction space, continuity there with its normalizations and cutoffs retained, and infinite iteration remain separate requirements.

4 Total Ward extraction through field degree four

The following argument completes the low-degree *total-kernel* classification. It does not impose gauge invariance on individual activity coefficients. Its exact statement concerns subsequential infinite-lattice kernels; finite-period winding terms and localization of the canceled functional remain separate questions.

4.1 Hypotheses and the reference action

For a compact connected simple matrix group, use Hermitian Lie-algebra coordinates and a fixed invariant positive trace form. Consider analytic total corrections E_N on exhausting compatible cubic tori. Assume the local gauge identity, output-unit translations, full signed permutations and reality on real fields established for the actual local sector in Section 1. Assume connected activity representations on a fixed cell partition with

$$\sup_I \sum_{H \ni I} e^{a_0|H|} \|E_{N,H}\|_b \le \epsilon. \tag{35}$$

Here $a_0, b > 0$, the cell size J, and the scalar coordinate count D per cell are fixed uniformly in volume and preceding refinement. The total symmetries need not hold for the chosen coefficients.

Let S_N be a real analytic, exactly gauge-invariant and spatially invariant reference, with $S_N(0) = DS_N(0) = 0$, and with uniform fixed-order kernel moment bounds through degree four. Assume its limiting midpoint Hessian has the normalization

$$\Gamma_2^S(p) = |p|^2 I - pp^T + O(|p|^4). \tag{36}$$

The normalized Wilson action is one choice. The actual homogeneous classical action S_{MR} is another: its Hessian is q_{MR}, whose uniform normalization was proved by the alias formula in Notebook07, and the classical strict-support construction supplies the higher fixed-order bounds. Ellipticity alone would not establish (36).

Theorem 4.1 (Full marginal completion from the total Ward identity)**.** *Under these assumptions, choose a joint subsequential infinite-lattice limit of the kernels of E_N, S_N through degree four. There is*

a real coefficient β such that the midpoint Fourier vertices of the residual $F = E - \beta S$ satisfy

$$\begin{aligned}\Gamma_2^F(p) &= O(|p|^4),\\ \Gamma_3^F(p_1,p_2,p_3) &= O\Big(\max_j |p_j|^3\Big),\\ \Gamma_4^F(p_1,p_2,p_3,p_4) &= O\Big(\max_j |p_j|^2\Big), \qquad \sum_j p_j = 0.\end{aligned} \tag{37}$$

Thus the total Taylor coefficients of field degree $n = 2, 3, 4$ and momentum degree at most $4 - n$ agree with βS. The one-point kernel vanishes; the vacuum is unrestricted.

For a semisimple product there is one coefficient per simple ideal. The analogous statement subtracts gauge-invariant references with the individual normalized Maxwell blocks. A single-coupling rebase is immediate for a simple group, or if those coefficients coincide. Reality can be omitted if a complex coefficient β is permitted.

4.2 Fixed-order kernels and their joint limit

Write $K_n = D^n E_N(0)$. Coordinate Cauchy on one coefficient gives

$$\sum_{i_1,\dots,i_n \in H} |D^n E_{N,H}(0)[e_{i_1},\dots,e_{i_n}]| \le (D|H| + n)^n b^{-n} \|E_{N,H}\|_b. \tag{38}$$

Indeed the exact monomial count multiplied by $n!$ is $n!\binom{D|H|+n-1}{n}$, bounded by the displayed factor. Fixing the first coordinate only decreases the sum. The graph Steiner length of slots contained in connected H is at most $|H| - 1$. For $0 < \sigma < a_0$, therefore,

$$\sup_{i_1} \sum_{i_2,\dots,i_n} e^{\sigma\tau(i_1,\dots,i_n)} |K_n(i_1,\dots,i_n)| \le b^{-n} \epsilon \sup_{m \ge 1} (Dm + n)^n e^{-(a_0-\sigma)m}. \tag{39}$$

Scalar and color indices are included in the sums. A further fixed exponent loss absorbs every fixed spatial moment. Conversion from the J-cell metric to the unit-site metric retains J-dependent constants.

Global constant gauges imply $K_1 = 0$ separately at each bond, because its covector annihilates every Lie commutator. Translation invariance permits an anchor at the origin. The bounds then give a diagonal subsequence of all fixed displacements, simultaneously for $n \le 4$ and for the reference. Fatou preserves exponential summability. Permutation symmetry of the slots, signed permutations, translation and global adjoint covariance pass to this limit. The differentiated Ward identities below pass as well: their coefficient equations contain finite nearest-neighbor shifts and lower kernels with coinciding slots. Equivalently one may first test against finitely supported fields and gauges.

The limiting Fourier vertices are analytic near the simultaneous origin, and their fixed derivatives are justified by the spatial moments. No unnormalized infinite-volume total functional is being summed; even a nonzero vacuum density would preclude that.

4.3 The exact logarithmic gauge generator

For $U_b = e^{iA_b}$, use

$$U_b(t) = e^{it\lambda_{b-}} U_b e^{-it\lambda_{b+}}, \qquad (d\lambda)_b = \lambda_{b_+} - \lambda_{b_-}.$$

Differentiating the exponential chart, with $\mathrm{ad}_A X = [A, X]$, gives

$$\delta_\lambda A = i[\lambda_-, A] - f(i\,\mathrm{ad}_A) d\lambda, \qquad f(z) = \frac{z}{1 - e^{-z}} = 1 + \frac{z}{2} + \frac{z^2}{12} + O(z^4). \tag{40}$$

In particular

$$\begin{aligned}\delta_\lambda A &= -d\lambda + L_\lambda(A) + Q_\lambda(A,A) + O(A^4\lambda),\\ L_\lambda(A)_b &= \frac{i}{2}[\lambda_{b_-} + \lambda_{b_+}, A_b],\\ Q_\lambda(A,B)_b &= \frac{1}{24}([A_b,[B_b,d\lambda_b]] + [B_b,[A_b,d\lambda_b]]).\end{aligned} \tag{41}$$

The cubic Bernoulli coefficient is zero. Comparing powers in $DE(A)[\delta_\lambda A] = 0$, using $K_1 = 0$, gives

$$\begin{aligned}K_2(A,d\lambda) &= 0,\\ K_3(A,A,d\lambda) &= 2K_2(A,L_\lambda(A)),\\ K_4(A,A,A,d\lambda) &= 3K_3(A,A,L_\lambda(A)) + 6K_2(A,Q_\lambda(A,A)).\end{aligned} \tag{42}$$

Polarization yields

$$K_3(A,B,d\lambda) = K_2(A,L_\lambda(B)) + K_2(B,L_\lambda(A)), \tag{43}$$

$$\begin{aligned}K_4(A,B,C,d\lambda) = {}& K_3(A,B,L_\lambda(C)) + K_3(A,C,L_\lambda(B)) + K_3(B,C,L_\lambda(A))\\ &+ 2K_2(A,Q_\lambda(B,C)) + 2K_2(B,Q_\lambda(A,C)) + 2K_2(C,Q_\lambda(A,B)).\end{aligned} \tag{44}$$

These are total-kernel identities. Subtracting just a quadratic polynomial would generally violate them. Subtracting a multiple of an exactly invariant reference preserves them.

The generator and differentiated identities are independently derived here. Their published counterparts are Bałaban, CMP 109 (1987), (4.9)–(4.15), pp. 283–284; the full-action subtraction is (4.38)–(4.45), pp. 291–292. The stronger coefficientwise orbit hypotheses of that article are not assumed.

4.4 Midpoint phases and parity

Use bond-midpoint Fourier fields and site Fourier gauges:

$$A_\mu(x) = \int e^{ip\cdot(x+e_\mu/2)} a_\mu(p)\,dp, \qquad \lambda(x) = \int e^{ik\cdot x}\lambda(k)\,dk.$$

On $\sum_j p_j = 0$, anchoring the first bond gives the vertex phase

$$\exp\left\{ i\sum_{j=2}^{n} p_j\cdot(x_j + e_{\mu_j}/2 - e_{\mu_1}/2)\right\}. \tag{45}$$

Only local analyticity near zero momentum is required; midpoint vertices need not be individually periodic under every 2π shift. The Fourier coefficients in (41) are exactly

$$\begin{aligned}(d\lambda)_\mu :&\quad 2i\sin(k_\mu/2)\lambda(k),\\ L_\lambda(A)_\mu :&\quad i\cos(k_\mu/2)[\lambda(k), a_\mu(p)],\\ Q_\lambda(A,B)_\mu :&\quad \frac{i}{12}\sin(k_\mu/2)([a_\mu(p),[b_\mu(q),\lambda(k)]] + [b_\mu(q),[a_\mu(p),\lambda(k)]]).\end{aligned} \tag{46}$$

Their output momenta are $k, p+k, p+q+k$. Thus L has a bounded analytic coefficient, while Q supplies one power of gauge momentum.

Full inversion sends $a_\mu(p)$ to $-a_\mu(-p)$. Consequently

$$\Gamma_n(-p_1,\dots,-p_n) = (-1)^n\Gamma_n(p_1,\dots,p_n). \tag{47}$$

The cubic vertex is odd; the quadratic and quartic vertices are even. Midpoint phases are essential for this form.

4.5 Proof of the marginal completion

Proof of Theorem 4.1. Theorem 1.2 gives

$$\Gamma_2^E(p) = \beta(|p|^2 I - pp^T) + O(|p|^4).$$

For completeness, the continuous limiting Ward relation $\Gamma_2(p)\widehat{p} = 0$ first implies $\Gamma_2(0) = 0$ by taking $p = te_\mu$. Reflections and permutations restrict the degree-two tensor to diagonal $Ap_\mu^2 + B\sum_{\nu\neq\mu} p_\nu^2$ and off-diagonal $Cp_\mu p_\nu$. Transversality gives $A = 0, C = -B$. Put $\beta = B$, which is real under the stated reality assumption. It can be extracted as

$$\beta = \frac{1}{2}\,\partial_{p_\nu}^2 (\Gamma_2^E)_{\mu\mu}(0), \qquad \mu \neq \nu, \tag{48}$$

in a trace-orthonormal color direction. The reference normalization proves $\Gamma_2^F(p) = O(|p|^4)$.

Apply (43) to the residual and momenta tp, tq, tk, with $p + q + k = 0$. Its right side is $O(t^4)$: each residual Hessian is evaluated at small momentum, and the L coefficient in (46) is bounded. Parity gives

$$\Gamma_3^F(tp, tq, tk) = tP_1(p, q, k) + O(t^3).$$

The left side contracts its last vector index with $2i\sin(tk/2) = itk + O(t^3)$. Its degree-two coefficient therefore vanishes. Slot permutation symmetry gives, for each choice of colors,

$$p_\mu (P_1)_{\mu\nu\rho} = 0, \qquad q_\nu (P_1)_{\mu\nu\rho} = 0, \qquad k_\rho (P_1)_{\mu\nu\rho} = 0. \tag{49}$$

Eliminate $k = -p - q$ and write

$$(P_1)_{\mu\nu\rho} = \sum_\sigma A_{\mu\nu\rho;\sigma} p_\sigma + B_{\mu\nu\rho;\sigma} q_\sigma.$$

In the first relation of (49), the independent mixed monomials $p_\mu q_\sigma$ force every $B_{\mu\nu\rho;\sigma}$ to vanish. The second relation then forces every $A_{\mu\nu\rho;\sigma}$ to vanish by the independent monomials $q_\nu p_\sigma$. Thus $P_1 = 0$, proving the cubic remainder. This elementary argument permits arbitrary complex/color coefficients and does not presume a separately invariant cubic functional.

Finally apply (44), scaling all four momenta by t. Its K_3, L terms are $O(t^3)$; its K_2, Q terms are $O(t^5)$, because Q has the sine factor. By parity write

$$\Gamma_4^F(tp_1, \ldots, tp_4) = P_0 + O(t^2).$$

The order-t term on the left contracts P_0 with the last momentum p_4. That momentum is arbitrary: the other three can always sum to $-p_4$. Hence every component of P_0 vanishes. This proves the quartic remainder and completes the theorem. □

The argument is an identity of the integrated vertices. It does not choose a unique density modulo lattice divergences.

4.6 Reference independence and the full action

For the standard Wilson reference, the marginal continuum polynomial is explicit. With Hermitian coordinates set

$$f_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad F_{\mu\nu} = f_{\mu\nu} + i[A_\mu, A_\nu].$$

Its engineering-degree-four part is

$$\frac{1}{2}\sum_{\mu<\nu} \tau(F_{\mu\nu}^2) = \frac{1}{2}\sum_{\mu<\nu} \tau(f_{\mu\nu}^2) + i\sum_{\mu<\nu} \tau(f_{\mu\nu}[A_\mu, A_\nu]) - \frac{1}{2}\sum_{\mu<\nu} \tau([A_\mu, A_\nu]^2). \tag{50}$$

Indeed bond angles tA evaluated at midpoints of a lattice of spacing t give the plaquette $P = I + it^2 F_{\mu\nu} + O(t^3)$. Therefore $1 - \frac{1}{2}\tau(P + P^{-1}) = \frac{1}{2}t^4\tau(F_{\mu\nu}^2) + O(t^5)$. This verifies the normalization and the relative cubic and quartic coefficients.

No explicit Wilson expansion is needed in the theorem. Applying it to the difference of two exactly invariant normalized references shows that their marginal total vertices agree. In particular one may subtract the actual homogeneous classical reference directly:

$$e^{-g^{-2}S_{MR}+E} = e^{-(g^{-2}-\beta)S_{MR}+(E-\beta S_{MR})}. \tag{51}$$

The vacuum is separate. This identity neither proves positivity of $g^{-2} - \beta$ nor the required sign, size or accumulated drift of its actual increments.

4.7 Moment remainders and the remaining norm question

For the residual anchored kernels use the *bond midpoints* $\xi_j = x_j + e_{\mu_j}/2$, and define

$$T_{n,m} = \sup_{i_1} \sum_{i_2,\dots,i_n} \left(\sum_{j=2}^{n} |\xi_j - \xi_1|_1 \right)^m |K_n^F(i_1, \dots, i_n)|. \tag{52}$$

They are bounded by the moments of E plus $|\beta|$ times those of S. The phase (45) and the real one-variable Taylor integral remainder give componentwise

$$\begin{aligned} |\Gamma_2^F(p)| &\le \frac{T_{2,4}}{4!}\|p\|_\infty^4, \\ |\Gamma_3^F(p_1,p_2,p_3)| &\le \frac{T_{3,3}}{3!}\max_{j\ge 2}\|p_j\|_\infty^3, \\ |\Gamma_4^F(p_1,p_2,p_3,p_4)| &\le \frac{T_{4,2}}{2!}\max_{j\ge 2}\|p_j\|_\infty^2. \end{aligned} \tag{53}$$

Fixed vector/color row sums add only their fixed dimensions. The exponential kernel bound makes these moments finite.

The cancellations also hold in ordinary tail-anchored coordinates: conversion multiplies each midpoint component by a nonzero analytic phase and cannot lower its vanishing order. Thus all residual ordinary momentum jets through degrees $3, 2, 1$ vanish for field degrees $2, 3, 4$, respectively. This includes the weaker anchored moments used in second-order quadratic, first-order cubic and zeroth-order quartic spatial subtractions.

The constants can depend strongly on the fixed old grouping J. Since the common-domain construction chooses J after R, an estimate $C(J)R^{-\alpha}$ is not automatically a contraction. No such claim is made. The remaining tasks are localization of the canceled total without false coefficientwise Ward assumptions, finite-period wrapping control or a consistent infinite correction, a spatial-gain norm with the correct parameter order, and the actual relevant/marginal recursion. The theorem proves the total-kernel Taylor classification; it does not supply these separate conclusions.

5 Classical background composition modulo gauge

Fix the primitive odd ratio ℓ and write $M = \ell^k$, $R_1 = \ell^{s_1}$ and $R_2 = \ell^{s_2}$. Let $U_n(D)$ be the selected ordinary homogeneous Wilson minimizer at refinement n. The ordinary-minimum theorem of Notebook15 gives a refinement-uniform small global $C^{1,\alpha}$ comparison ball, a bound $\|A_{\mathrm{o},n}(D)\|_{C^{1,\alpha}} \le C\|\log D/i\|_\infty$, and gauge equivalence whenever equality holds in that comparison class.

Define the pre-root-gauge map and its scalar substitution by

$$P_{M,R}(D) = M_\ell^k(U_{MR}(D)), \qquad (\mathcal{T}_{M,R}E)(D) = E(P_{M,R}(D)). \tag{54}$$

Here $R = \ell^s$ and $M_\ell^s P_{M,R}(D) = D$. All expressions include the natural identification of the nested lattices. The input D lives on the next-coarse lattice and $P_{M,R}(D)$ on its R-times finer current lattice. The prefix estimate gives a common classical radius b_{cl} and constant C_0, independent of M, R and compatible volume, such that

$$\left\| \frac{1}{i} \log P_{M,R}(e^{iw}) \right\|_\infty \le \frac{C_0}{R} \|w\|_\infty. \tag{55}$$

The operator $\mathcal{T}$ is a classical background substitution, not a fluctuation integral.

5.1 Two minimum comparisons on the same fine graph

For a final datum Z put

$$U_2 = U_{MR_1R_2}(Z), \qquad D_1 = M_\ell^{k+s_1}(U_2) = P_{MR_1,R_2}(Z), \qquad U_1 = U_{MR_1}(D_1). \tag{56}$$

Both fine fields live on the same original oriented graph. In final-Z units its spacing is $\eta_2 = (MR_1R_2)^{-1}$ and the D_1 spacing is R_2^{-1}. The definition of $U_{MR_1}(D_1)$ instead uses D_1-unit coordinates and spacing $\eta_1 = (MR_1)^{-1} = R_2\eta_2$. This is a coordinate relabeling of the same links. In four dimensions the coupling-free Wilson action is the same plaquette sum in both conventions.

Exact ordinary composition gives

$$\begin{aligned} M_\ell^{k+s_1}(U_1) &= D_1 = M_\ell^{k+s_1}(U_2), \\ M_\ell^{k+s_1+s_2}(U_1) &= Z = M_\ell^{k+s_1+s_2}(U_2). \end{aligned} \tag{57}$$

For fixed finite R_1, R_2 both fields lie in the needed comparison balls after shrinking the real Z radius, uniformly in M and volume. Explicitly, a fine physical potential A_1 in D_1 units becomes

$$A_1^Z(x) = R_2 A_1(R_2 x), \qquad \|A_1^Z\|_{C^{1,\alpha}} \le C R_2^{2+\alpha} \|A_1\|_{C^{1,\alpha}}. \tag{58}$$

The reverse conversion decreases the norm up to a fixed convention constant, and decreases its supremum by R_2^{-1}. By (55), $\|\log D_1/i\| \le C_0 \|\log Z/i\|/R_2$. Thus U_1 in final units is bounded by $CR_2^{1+\alpha}\|\log Z/i\|$, while U_2 in D_1 units is bounded by $C\|\log Z/i\|/R_2$. These are finitely many radius restrictions for fixed ratios.

The minimum at D_1, with competitor U_2, and the minimum at Z, with competitor U_1, yield respectively

$$S(U_1) \le S(U_2), \qquad S(U_2) \le S(U_1). \tag{59}$$

Equality and the ordinary-minimum equality clause give a fine gauge g with $U_1 = U_2^g$. This gauge is supplied by the small ordinary-to-corrected transfer, the constraint-preserving flow into the flat slice, and the inverse transfer from the same strict slice minimizer. These controlled gauge transformations can be joined to the identity inside a smaller common neighborhood. Equal action alone, without that equality clause, would not justify this conclusion.

Ordinary gauge covariance gives

$$M_\ell^k(U_1) = (M_\ell^k(U_2))^{g|_{\mathrm{current}}}. \tag{60}$$

Consequently

$$P_{M,R_1}(P_{MR_1,R_2}(Z)) \quad \text{is gauge equivalent to} \quad P_{M,R_1R_2}(Z). \tag{61}$$

The two selected representatives need not coincide. The connecting gauge need not be unique; its restriction to D_1 sites stabilizes D_1.

5.2 The scalar germ identity and its continuation

Theorem 5.1 (Classical background semigroup modulo gauge)**.** *Let E be a scalar total functional holomorphic on a current-angle product ball and locally gauge invariant there. On finite compatible regulators,*

$$\mathcal{T}_{MR_1,R_2}(\mathcal{T}_{M,R_1}E) = \mathcal{T}_{M,R_1R_2}E \tag{62}$$

as holomorphic germs at zero. The identity extends to every connected common product domain of the specified branches. The output $\mathcal{T}_{M,R}E$ is locally gauge invariant. In particular, for any fixed finite q,

$$\mathcal{T}_{MR^{q-1},R}\cdots\mathcal{T}_{MR,R}\mathcal{T}_{M,R}E = \mathcal{T}_{M,R^q}E. \tag{63}$$

If $R > C_0$ and b lies in the common classical and original E domains, both sides of (63) are analytic on the same radius-b ball and agree there.

Proof. On a sufficiently small real Z ball, the gauge path in (60) remains inside the original domain of E. Total gauge invariance proves (62). Both compositions are holomorphic; uniqueness from the real slice proves the germ identity and its continuation. No gauge identity for separate activity coefficients is used.

To see local invariance of the output, extend a small coarse gauge smoothly to prescribed fine point values. The transformed minimum at D is a small feasible competitor at D^v. Compare with the minimum at D^v, and perform the inverse-gauge comparison at D. Equality follows; the ordinary-minimum equality clause identifies the two configurations modulo fine gauge. Their partial averages are gauge equivalent, so evaluating E gives local invariance. Analyticity continues this identity from the real overlap.

The identity can now be iterated, proving (63). If each ratio is larger than C_0, (55) maps the radius-b ball into itself at every substitution; the direct map has the same property. Thus both sides are analytic on the connected common b ball. Their germ equality continues there, even if the real variational comparison required a smaller radius depending on the finite ratios or on q. This uses no complex minimum assertion. □

5.3 Use and limitations of a finite iterate

The direct representative in (63) obeys the Hölder prefix estimates

$$\|a_{M,R^q}\|_\infty \le CbR^{-q}, \qquad \|\Delta a_{M,R^q}\|_\infty \le CbR^{-2q}, \qquad \|\Delta^2 a_{M,R^q}\|_\infty \le CbR^{-(2+\alpha)q}, \tag{64}$$

with primitive constants independent of q. Its complete supported families follow by applying the same prefix theorem at endpoint ratio R^q.

This permits keeping the elementary R and grouping J fixed before selecting a finite q. A separately proved bound $C(J)R^{-q\alpha}$ could then have a noncircular parameter order. Such an activity-norm estimate is not proved by the scalar identity alone.

The canonical families from successive core/tail transports need not equal the direct family. Their scalar sums agree; a new justified localization may represent that common scalar function. No automatic bound for the old sequential family, or bounded relocalization on a chosen quotient norm, follows.

Nor does (63) identify sequential quantum integrals with one integral at ratio R^q. Sequential steps retain their separate gauge factors, normalizations, windows and newly generated interactions. Those terms do not disappear from an iterated quantum recursion. This is a finite classical background semigroup modulo gauge, uniform in preceding refinement and compatible volume; infinite iteration and regulator consistency remain open.

6 Connected localization of spatially canceled low kernels

This section proves a connected-activity norm estimate for the canceled *total* kernels. Its inputs are coherent complete-field support families and exact infinite-lattice moment cancellations. It also applies to explicitly periodized versions of those inputs, with their lifted sums. It does not silently impose lifted moment identities on an arbitrary finite torus kernel.

6.1 Inputs and parameter order

Work in dimension four. A current coordinate consists of a site, positive bond direction and color, with fixed internal dimension v. For $n = 2, 3, 4$, let K_n be a translation-invariant symmetric n-linear kernel. Anchor its first site at x, write the other sites as $x + r_2, \dots, x + r_n$, and put $m = \sum_{j=2}^{n} |r_j|_1$. Internal indices are included in all sums. Assume

$$\sup_{\text{first internal index}} \sum_{\text{other indices}, r_2, \dots, r_n} e^{\sigma m} |K_n(0, r_2, \dots, r_n)| \le \epsilon b^{-n}. \tag{65}$$

The constants σ, ϵ can incorporate the fixed old grouping J. Fixed-order Cauchy and a connected activity bound supply this estimate with a smaller spatial exponent.

Use tail-site coordinates. Assume componentwise

$$\begin{aligned} &\sum_{r} r^{\gamma} K_2(r) = 0 && (|\gamma| \le 3), \\ &\sum_{r,s} K_3(r, s) = 0, \qquad \sum_{r,s} r_\rho K_3(r, s) = \sum_{r,s} s_\rho K_3(r, s) = 0 && (\rho = 1, \dots, 4), \\ &\sum_{r,s,t} K_4(r, s, t) = 0. \end{aligned} \tag{66}$$

These are consequences of the stronger vanishing orders in Theorem 4.1; analytic conversion from midpoint to tail coordinates preserves those orders.

Fix an integer endpoint ratio L, put $h = L^{-1}$, and let $a(w)$ be an analytic current-angle background on the radius-b final-field ball. Current sites have physical spacing h. Assume coherent complete-field strict families: on every fixed enlarged final-unit collar rooted at i,

$$a|_i = \sum_{X \ne \varnothing} a_{i,X}(w_X), \qquad a_{i,X}(0) = 0, \qquad \sup_i \sum_X e^{\kappa\tau(\{i\} \cup X) + A|X|} \|a_{i,X}\|_{C^{1,\alpha};b} \le C_* b h. \tag{67}$$

The derivatives in this norm are physical. Coefficients on overlapping collars restrict one global assignment. All maxima, first derivatives and first-gradient Hölder quotients are inside the support sum. A fixed collar contains any coordinate paths whose current length is at most L; enlarging it changes only fixed constants and possibly decreases κ. These are the complete-family inputs of the supported pre-gauge background construction, not bounds merely on the summed field.

Let $P_n(a) = K_n(a, \dots, a)/n!$, fix $a_0 > 0$, and set $c = a_0 + 1$. Choose J once with

$$\frac{2c}{J} \le \kappa, \qquad A \ge 4c. \tag{68}$$

Theorem 6.1 (Low-kernel spatial transport)**.** *There are C, L_0, depending on $\sigma, J, C_*, a_0, \alpha$ and the fixed internal dimension, but not on L, b, volume or preceding refinement, such that for $L \ge L_0$ there are connected activities on final J-cells with*

$$\sum_H B_H(w_H) = \sum_{n=2}^{4} P_n(a(w)), \qquad \sup_I \sum_{H \ni I} e^{a_0|H|} \|B_H\|_b \le C\epsilon L^{-\alpha}. \tag{69}$$

The construction is linear in the total kernels. Individual B_H need not be gauge invariant. The scalar identity is vacuum-normalized and absolutely convergent for compactly supported final fields.

The constant is fixed after J and before L. One can therefore set $L = R^q$ with fixed R, J and choose a finite q afterward. This statement does not identify q successive quantum integrations with one direct fluctuation integral.

6.2 A valid quadratic difference factorization

The matrix kernel need not be symmetric at each displacement, even though Hessian reciprocity gives $K_2(r)^T = K_2(-r)$. We use a Laurent factorization that does not require pointwise symmetry. Set

$$F(z) = \sum_r K_2(r) z^r, \qquad d_\rho^+(z) = z_\rho - 1, \qquad d_\rho^-(z) = z_\rho^{-1} - 1.$$

The exponential bound makes F holomorphic on a product annulus. Equation (66) says its jets through degree three at $z = (1, \dots, 1)$ vanish.

For a Laurent function define the ordered divided difference

$$\mathcal{B}_j F(z) = \frac{F(1, \dots, 1, z_j, \dots, z_4) - F(1, \dots, 1, 1, z_{j+1}, \dots, z_4)}{z_j - 1}. \tag{70}$$

Its singularity at $z_j = 1$ is removable, and

$$F(z) - F(1) = \sum_j (z_j - 1) \mathcal{B}_j F(z).$$

For a monomial z^r, the quotient in coordinate j has $|r_j|$ monomials, each at absolute exponent length at most $|r|_1$, including negative exponents. Therefore, for $0 < \sigma' < \sigma$, the combined absolute row Wiener norm obeys

$$\sup_u \sum_{j,v,r} e^{\sigma'|r|_1} |(\mathcal{B}_j F)_{uv}(r)| \le \frac{1}{e(\sigma - \sigma')} \sup_u \sum_{v,r} e^{\sigma|r|_1} |F_{uv}(r)|. \tag{71}$$

Here the j sum is inside the row supremum. The bound uses $\sum_j |r_j| = |r|_1$ and $\sup_{t \ge 0} t e^{-(\sigma - \sigma')t} = [e(\sigma - \sigma')]^{-1}$. The local Taylor series also shows that $\mathcal{B}_j$ lowers a vanishing order by at most one.

Apply the identity twice, using exponents $\sigma, 3\sigma/4, \sigma/2$. Put

$$H_{\rho\tau} = \mathcal{B}_\tau \mathcal{B}_\rho F, \qquad \widetilde{H}_{\rho\tau}(z) = -z_\rho H_{\rho\tau}(z).$$

Since $d_\rho^-(-z_\rho) = d_\rho^+$, this gives

$$F(z) = \sum_{\rho,\tau} d_\rho^-(z) \widetilde{H}_{\rho\tau}(z) d_\tau^+(z), \qquad \widetilde{H}_{\rho\tau}(1) = 0. \tag{72}$$

In fact the jets of $\widetilde{H}_{\rho\tau}$ through degree one vanish. Its combined kernel bound is

$$\sup_u \sum_{\rho,\tau,v,r} e^{(\sigma/2)|r|_1} |(\widetilde{H}_{\rho\tau})_{uv}(r)| \le \frac{16 e^{\sigma/2}}{e^2 \sigma^2} \epsilon b^{-2}. \tag{73}$$

The factor $e^{\sigma/2}$ pays the unit shift. No symmetry of the individual $\widetilde{H}_{\rho\tau}$ is asserted.

Let $D_\rho a(x) = a(x + e_\rho) - a(x)$, the unscaled current difference. Integration by parts transfers one difference to each field. The zero sum in (72) then gives the exact identity

$$P_2(a) = \frac{1}{2} \sum_{x,r,\rho,\tau} [D_\rho a(x)]^T \widetilde{H}_{\rho\tau}(r) [D_\tau a(x + r) - D_\tau a(x)]. \tag{74}$$

The sign follows because d_ρ^- is the bilinear adjoint of d_ρ^+. Compactly supported fields give the identity directly; the absolute majorants below extend it to the substituted fields. No positivity or second field derivative is used.

6.3 Cubic and quartic cancellation

Put

$$\ell_r(x) = \sum_\rho r_\rho D_\rho a(x), \qquad R_r(x) = a(x+r) - a(x) - \ell_r(x). \tag{75}$$

In the anchored cubic expression

$$P_3(a) = \frac{1}{6} \sum_{x,r,s} K_3(r,s)[a(x), a(x+r), a(x+s)],$$

replace each shifted factor by $a(x) + \ell + R$. The zeroth moment cancels the all-anchor term. The first moments cancel a, ℓ, a and a, a, ℓ. Six terms remain: the two with one R and no ℓ, the ℓ, ℓ term, the two mixed ℓ, R terms, and the R, R term, always with unshifted first factor.

For the quartic expression subtract the all-anchor product using its zeroth moment, then telescope the three shifted factors. Each of three terms has one difference $a(x + r_j) - a(x)$ and three endpoint or anchor factors. All cancellations are made in the total kernels before triangle inequalities or final carrier assignment.

6.4 Near-displacement coefficient bounds

Split these exact residual sums into $m \le L$ and $m > L$. Neither restricted kernel is claimed to retain zero moments. For the quadratic identity use $m = |r|_1$. In the near case all evaluations and coordinate paths lie in a fixed final-unit collar. Coherence in (67) gives rooted coefficient-family bounds

$$\begin{aligned} |a| &\le Cbh, \qquad |\ell_r| \le Cbh^2|r|_1, \\ |a(x+r) - a(x)| &\le Cbh^2|r|_1, \qquad |R_r| \le Cbh^{2+\alpha}|r|_1^{1+\alpha}. \end{aligned} \tag{76}$$

For a negative coordinate traversal, the positive difference is evaluated at the lower endpoint with the traversal sign. Subtracting its value at the anchor and using the physical gradient Hölder bound proves the last estimate. Each inequality is for the weighted sum of coefficient norms, with its suprema and Hölder quotients inside that sum. It is not deduced from a bound on the scalar sum.

All near factors can be rooted at the anchor cell. Multiplication glues rooted trees and support unions, so their weights are submultiplicative. In the quadratic identity,

$$|D_\rho a(x)| \le Cbh^2, \qquad |D_\tau a(x+r) - D_\tau a(x)| \le Cbh^{2+\alpha}|r|_1^\alpha.$$

Its product is therefore

$$Cb^2h^{4+\alpha}|r|_1^\alpha. \tag{77}$$

Sum it with (73). The six cubic products are bounded by

$$Cb^3h^{4+\alpha}(1+m)^6. \tag{78}$$

The leading a, a, R term has that power of h; a, ℓ, ℓ has h^5, and the others have still higher powers. The quartic telescoping is bounded by

$$Cb^4h^5(1+m). \tag{79}$$

The weighted kernel sums cancel b^n and pay all fixed displacement moments. Per original current anchor, the three bounds are respectively $C\epsilon h^{4+\alpha}$, $C\epsilon h^{4+\alpha}$, and $C\epsilon h^5$.

6.5 Connected carriers and the anchor count

For each product coefficient keep its final-unit anchor i_0 and the endpoint roots actually used. In a near term the complete collar permits the common root i_0. In a far term use the endpoint roots and any neighboring roots needed for Da. Their number is at most a fixed n_0, independent of L and the displacement.

Project the finite root set and, for each field-factor occurrence, the terminal set consisting of its root and variable support X to final J-cells. For each projected terminal set choose a deterministic short macro connecting graph supplied by the projection lemma. Take the connected union of these macro graphs as H, including the anchor. The fine Steiner lengths below bound those chosen macro graphs; the carrier need not contain the full projection of an arbitrary fine connecting tree. The projection bound gives

$$\begin{aligned} &|H| \le 1 + T, \\ &T \le \frac{2}{J}\left[\tau(\text{roots}) + \sum_{\text{occ}} \tau(\{\text{root}\} \cup X)\right] + 4\left[n_0 + \sum_{\text{occ}} |X|\right]. \end{aligned} \tag{80}$$

Let $j(x)$ be the final J-cell containing x/L. For $z \in H$, its graph distance to $j(x)$ is at most T. Hence

$$e^{a_0|H|} 1_{\{z\in H\}} \le e^{a_0} e^{-\operatorname{dist}(z,j(x))} e^{cT}. \tag{81}$$

The budgets (68) pay every attached tree and support-cardinality factor. The fixed e^{4cn_0} is absorbed into C; near root-joining trees have bounded length.

There are exactly $(JL)^4$ current anchor sites per final J-cell, with the same centered-boundary convention throughout. Therefore

$$\sum_{x\in\mathbb{Z}^4} e^{-\operatorname{dist}(z,j(x))} \le (JL)^4 S_1, \qquad S_1 = \sum_{y\in\mathbb{Z}^4} e^{-|y|_1}. \tag{82}$$

The first internal-index count is the fixed v. There is no additional count of all output coordinates. The near contributions are consequently $CJ^4\epsilon L^{-\alpha}$ for degrees two and three, and $CJ^4\epsilon L^{-1}$ for degree four.

6.6 Far displacements and convergence

For $m > L$, use the same exact residual formulas but root each endpoint separately. The cubic and quartic factors obey

$$|a| \le Cbh, \qquad |\ell_r| \le Cbh^2 m, \qquad |R_r| \le Cbh(1 + hm).$$

Their products are bounded by $Cb^3h^3(1+m)^2$ and Cb^4h^4. For the quadratic expression both first differences are bounded by Cbh^2 at their own roots, so its product is bounded by Cb^2h^4.

The finite endpoint-root tree has

$$\tau(\text{roots}) \le C_d(m/L + 1).$$

After (81) this costs at most $C\exp(C'm/(JL))$. Use the common kernel exponent $\sigma_* = \sigma/2$, supplied by (73) and also available for K_3, K_4. Choose L_0 so that $C'/(JL) \le \sigma_*/4$. One half of σ_* pays the extra tree weight and the fixed polynomial, leaving an exponentially small tail. After (82) the total far bound is

$$CJ^4\epsilon(L+2)e^{-\sigma_* L/4} \le C'\epsilon L^{-\alpha}, \qquad L \ge L_0. \tag{83}$$

All constants here are fixed before L.

The absolute majorants justify regrouping by connected H and give holomorphic coefficients on the same open b-ball, with their bounded supremum norm. For compactly supported w, each nonzero field-factor coefficient has variable support meeting that compact set, since it vanishes at zero. Its final carrier contains that support. The anchored bound therefore implies convergence of the total sum, bounded by a constant times the number of final J-cells meeting the support. For that compact support K, every nonzero field coefficient also has $\tau(\{i\} \cup X) \geq \operatorname{dist}(i, K)$. Equation (67) therefore gives exponential spatial decay of $a(w)$ and its first differences. The original quadratic kernel sum is absolutely convergent as well, so truncation and exhaustion justify its integration by parts without a boundary term. The near and far bounds prove Theorem 6.1.

6.7 Scope of the relocalization

The result is a bounded connected localization of the canceled low-degree *total* functional. It is stronger than extensive scalar power counting and needs no invariant individual carriers. For a residual analytic activity $E - \beta S$, its coefficientwise degree-≥ 5 part can be transported by the previous high-jet estimate, while the total degrees two through four use the present construction. The two localizations may change the representation.

This section does not prove canonical finite-torus wrapping control, an infinite-volume quantum measure, equality of sequential and direct cutoff integrals, boundedness of a complete quantum map on a quotient of representations, or a coupling trajectory. The classical scalar semigroup permits the choice $L = R^q$ for a gauge-invariant total, but removes none of those separate interfaces.

7 A Banach quotient for analytic interactions

Fix the cell size J, a bound D on the number of scalar field coordinates per cell, a cardinality exponent $a > 0$, and a radius $b > 0$. Let $\mathcal{C}$ be the countable set of nonempty finite connected cell sets. A declared carrier may exceed the variables actually read by its coefficient. Let $H_b^\infty(H)$ denote the bounded holomorphic functions on the open scalar-coordinate radius-b polydisc of H. Define

$$\mathcal{A}_{a,b} = \left\{ E = (E_H)_{H \in \mathcal{C}} : \|E\|_{a,b} := \sup_i \sum_{H \ni i} e^{a|H|} \|E_H\|_b < \infty \right\}. \tag{84}$$

Individual coefficients need not have any symmetry. Constants are allowed. All constants below may depend on the fixed J, D, a.

7.1 Closed equality kernel and summable fields

For a finitely supported field A, $\|A\|_\infty < b$, put

$$\Phi_E(A) = \sum_H \big[E_H(A|_H) - E_H(0) \big]. \tag{85}$$

If K is its finite cell support, only carriers meeting K contribute, and

$$\sum_H |E_H(A|_H) - E_H(0)| \leq 2|K| \|E\|_{a,b}. \tag{86}$$

The series is normally convergent on every finite-coordinate polydisc.

Proposition 7.1 (Complete representation quotient)**.** *The space $\mathcal{A}_{a,b}$ is Banach. The subspace*

$$\mathcal{N}_{a,b} = \{ E : \Phi_E(A) = 0 \text{ for every compact } A,\ \|A\|_\infty < b \} \tag{87}$$

is closed, and

$$\mathcal{Q}_{a,b} = \mathcal{A}_{a,b}/\mathcal{N}_{a,b}, \qquad \|[E]\|_{\mathcal{Q}} = \inf_{F\in\mathcal{N}_{a,b}} \|E+F\|_{a,b} \tag{88}$$

is a Banach space. Its invariant subspace, defined by translation, hypercubic, global-adjoint and local-gauge invariance of the normalized total, is closed.

Proof. A norm-Cauchy sequence is Cauchy in each $H_b^\infty(H)$: choose one root in H in (84). Component limits exist. Fatou over carriers, followed by the root supremum, proves both the norm bound and norm convergence. The null space is the intersection of the kernels of the bounded evaluations (86). The usual quotient-completeness argument uses a summably Cauchy subsequence and representatives with summable successive differences; it requires no minimizing representative. Each total symmetry is an equality of bounded evaluations, for all compact configurations and transforms in their common domain. Its preimage is closed and contains $\mathcal{N}_{a,b}$. Reality specifies a closed real form. □

The quotient removes all carrierwise constants and all other representations of the zero normalized total. It does not select a physical vacuum counterterm; a vacuum sector, if needed, stays separate. Equality of germs on every finite-coordinate subspace is equivalent to (87) by analytic uniqueness.

Use the cell norm $\|A\|_1 = \sum_i \|A|_i\|_\infty$. For $\|A\|_\infty, \|B\|_\infty \le \rho < b$, directional Cauchy on the convex segment gives

$$\begin{aligned} |E_H(A|_H) - E_H(B|_H)| &\le \frac{\|E_H\|_b}{b-\rho}\|(A-B)|_H\|_\infty \\ &\le \frac{\|E_H\|_b}{b-\rho}\sum_{i\in H}\|A_i - B_i\|_\infty. \end{aligned} \tag{89}$$

Use a strictly smaller directional circle first, then take its radius to the displayed margin. Summing at each cell proves

$$|\Phi_E(A) - \Phi_E(B)| \le \frac{\|E\|_{a,b}}{b-\rho}\|A-B\|_1. \tag{90}$$

Thus (85) extends by an absolutely convergent series to summable fields in every strict interior ball. Finite field truncations converge in (90). A null family remains zero on these fields. The extension therefore depends only on the quotient class, and the estimate holds with the quotient norm after taking the infimum over representatives.

7.2 A continuous total Maxwell coefficient

For each fixed order $n \le 4$, scalar Cauchy and the exact finite monomial count give a summable total derivative kernel. Indeed the ordered derivative sum on a carrier is at most $(D|H|+n)^n b^{-n}\|E_H\|_b$. Connectedness bounds every slot displacement by $c_d J|H|$. Spending half the carrier exponent on spatial distance and half on the fixed polynomial gives, for sufficiently small fixed $\sigma = \sigma(a,J) > 0$,

$$\sup_{i_1}\sum_{i_2,\dots,i_n} e^{\sigma\sum_{j\ge 2}|x_j - x_1|}|K_n(E;i_1,\dots,i_n)| \le C_n(a,D,J)b^{-n}\|E\|_{a,b}. \tag{91}$$

All scalar and color slots are included. Fixed moments follow with a further exponent loss. This is a fixed-order estimate, not a dimension-free absolute Taylor estimate at all orders.

Each kernel component is a derivative on a finite-coordinate subspace and hence depends only on $[E]$. Taking a representative infimum gives (91) with $\|[E]\|_{\mathcal{Q}}$. For a simple gauge group and an invariant total, define in the trace-normalized color component

$$\beta([E]) = \frac{1}{2}\partial_{p_\nu}^2 \Gamma^E_{2,\mu\mu}(0), \qquad \mu \ne \nu. \tag{92}$$

Use positive-bond midpoints $\xi(x,\mu) = x + e_\mu/2$. The total Ward algebra of Theorem 4.1 applies directly to these infinite kernels. It identifies (92) as the Maxwell coefficient and makes the distinct pair immaterial. No subsequence is needed once the infinite interaction is specified. For real interactions this coefficient is real.

It is a continuous linear functional on the invariant quotient:

$$|\beta([E])| \le C_\beta(a, D, J) b^{-2} \|[E]\|_{\mathcal{Q}}. \tag{93}$$

For example, a second derivative component is bounded by $2b^{-2}\|E_H\|_b$, there are at most $D|H|$ second slots, and their squared midpoint separation is at most $(c_d J|H|)^2$. Thus, up to a fixed color-normalization factor, one can take

$$C_\beta = D(c_d J)^2 \sup_{m \ge 1} m^3 e^{-am}. \tag{94}$$

The factor b^{-2} is essential. For a semisimple product use the finite vector of coefficients and one reference per simple ideal.

7.3 Full-reference extraction without a small-radius loss

Let $[S]$ be an invariant reference with $\beta([S]) = 1$. Assume a representation on one fixed radius ρ satisfies $\|S\|_{a,\rho} \le M_S$, uniformly in preceding refinement, and its total first derivative vanishes. This is the uniform fixed-radius classical support input; normalization of the Hessian alone does not imply it.

Define the homogeneous phase projections coefficientwise by

$$(P_n E_H)(A) = \frac{1}{2\pi} \int_0^{2\pi} e^{-int} E_H(e^{it} A)\, dt. \tag{95}$$

They retain H and have norm at most one. Replace S_H by $S_H^{\ge 2} = \sum_{n \ge 2} P_n S_H$. The normalized total has the same class, since constants are removed and the total linear sector is zero. Homogeneity yields, for $b \le \rho/2$,

$$\|S^{\ge 2}\|_{a,b} \le \frac{(b/\rho)^2}{1 - b/\rho} M_S \le C_S b^2, \qquad C_S = \frac{2M_S}{\rho^2}. \tag{96}$$

No vanishing of individual linear coefficients or individual gauge invariance was assumed. A joint subsequential limit preserves this bound by Fatou.

The full-reference projection

$$\mathcal{P}_S[E] = [E] - \beta([E])[S] \tag{97}$$

satisfies $\beta \mathcal{P}_S = 0$, $\mathcal{P}_S^2 = \mathcal{P}_S$, and

$$\|\mathcal{P}_S\| \le C_{\mathrm{ext}}, \qquad C_{\mathrm{ext}} = 1 + C_\beta C_S. \tag{98}$$

The constant is independent of small b. Apply (93) and (96) to representatives approaching the quotient infimum. The whole invariant reference is subtracted, so its nonlinear Ward identities remain exact.

7.4 Classical substitution on the quotient

Fix L and an actual infinite pre-gauge background map a_L. Assume its coherent whole-fiber support family satisfies

$$a_L|_i = \sum_{X \ne \varnothing} a_{i,X}(w_X), \qquad a_{i,X}(0) = 0, \qquad \sup_i \sum_X e^{\kappa\tau(\{i\} \cup X) + A|X|} \|a_{i,X}\|_b \le Cb/L, \tag{99}$$

and $\|a_L(w)\|_\infty \le \rho_0 < b$. The maximum over the whole current output fiber is inside the support sum, and overlapping collars use one coefficient assignment. For compact w supported in K, a nonzero term has $X \cap K \neq \varnothing$; hence

$$\|a_L(w)|_i\| \le (Cb/L)e^{-\kappa \operatorname{dist}(i,K)}. \tag{100}$$

At fixed L there are finitely many current coordinates per fiber, so $a_L(w)$ is summable. Consequently

$$(\mathcal{T}_L\Phi_E)(w) = \Phi_E(a_L(w)) \tag{101}$$

depends only on $[E]$, by (90). This is algebraic classical descent.

For bounded descent into an activity quotient, suppose a connected strict representation $\mathscr{L}_L E$ of (101) obeys

$$\|\mathscr{L}_L E\|_{a_{\mathrm{out}},b} \le B_L \|E\|_{a,b}. \tag{102}$$

Then $\mathcal{T}_L[E] = [\mathscr{L}_L E]$ is well defined and has norm at most B_L. Distinct valid output assignments have the same normalized total; equivalent inputs give equal output totals by summable-field continuity. The class map is linear because its scalar total is linear in E, even without a linear choice of representatives. Taking near-minimal input representatives proves the norm bound. Scalar amplitude alone does not prove (102).

The compactness construction of Section 3 provides simultaneous infinite interactions and coherent maps under its fixed-parameter finite-regulator hypotheses. Its summable-field composition limit transfers finite classical identities only on one jointly chosen system, with common strict domain margins. Unrelated subsequences cannot be composed. Preserving invariant totals needs more than an abstract orbit relation. The connecting infinite gauge must be one to which the input total's gauge identity applies. A sufficient condition is approximation by allowed compact gauges with transformed fields converging in ℓ^1 inside a common strict margin; then (90) passes the identity. A joint finite invariant approximation of a particular interaction and the maps is another sufficient condition for that interaction. For an action on the entire invariant quotient, this requirement must hold for every input class. A joint approximation of one input, or unrelated finite subsequences, does not establish a universal invariant-quotient action.

7.5 Conditional classical contraction after extraction

The following statement keeps the infinite-family and scale requirements explicit. Fix J, a and the primitive block ratio. Use one joint infinite system with normalized references S_M and S_{ML} satisfying (96) uniformly. Assume the coherent physical $C^{1,\alpha}$ families of Proposition 2.2, with primitive constants independent of L, M, and impose the fixed support reserves of Theorem 6.1 after choosing J. Assume also the core/tail budgets of the fifth-order background transport theorem of Notebook28, with unchanged a, b. Its constant C_{high} is independent of L once these reserves are fixed. Increasing L weakens its compression requirement; the same fixed J and logarithmic tail cutoff remain admissible. Choose b after the final finite L, so $b \le r_A/L_*$ with $L_* = L/(8C_A)$ and all other classical margins hold. Scalar substitution and preservation of the invariant domain satisfy the additional gauge-identity requirement just specified. An operator assertion on the whole invariant quotient assumes it for every class; a joint finite approximation may instead verify the following pointwise estimate on its specified collection of classes.

Theorem 7.2 (Classical quotient estimate)**.** *Under these hypotheses, in dimension four,*

$$\|\mathcal{T}_L\mathcal{P}_{S_M}[E]\|_{\mathcal{Q}} \le C_{\mathrm{ext,in}}\big[C_{\mathrm{low}}(J)L^{-\alpha} + 6C_{\mathrm{high}}L^{-1}\big]\|[E]\|_{\mathcal{Q}}. \tag{103}$$

If a normalized output reference is used, then

$$\|\mathcal{P}_{S_{ML}}\mathcal{T}_L\mathcal{P}_{S_M}\| \le C_{\mathrm{ext,out}}C_{\mathrm{ext,in}}\big[C_{\mathrm{low}}(J)L^{-\alpha} + 6C_{\mathrm{high}}L^{-1}\big]. \tag{104}$$

The constants are fixed after J and before L, b. A sufficiently large finite allowed $L = R^q$, followed by a sufficiently small common positive b, makes the last bound less than one.

Proof. Put $[F] = \mathcal{P}_{S_M}[E]$ and choose any representative. The phase projections (95) retain carriers, have norm at most one, and give a coefficientwise fifth-order remainder $R_5 = I - \sum_{n=0}^{4} P_n$ with norm at most six. These projections also descend to the unrestricted quotient: apply the phase integral to a zero normalized analytic total. They need not preserve gauge invariance separately.

The total linear term of F is zero by global adjoint invariance, its vacuum is removed, and $\beta(F) = 0$. The total Ward algebra therefore gives the low-kernel cancellations in Theorem 6.1. The kernel bound (91) supplies that theorem's ϵ with at most a fixed $C_{\mathrm{ker}}(J)\|F\|_{a,b}$. Its new connected representation has norm at most $C_{\mathrm{low}}(J)L^{-\alpha}\|F\|_{a,b}$. The coefficientwise R_5 part has a connected representation of norm at most $6C_{\mathrm{high}}L^{-1}\|F\|_{a,b}$ by the fifth-order transport theorem. The zero total linear term stays zero after substitution by (90); it need not vanish on each carrier.

All scalar decompositions are legitimate on summable inputs: the phase projections and R_5 lie in the anchored space, Section 7.1 gives their continuous extensions, and the new output families converge normally on compact fields. Adding the two representations and taking the infimum over representatives of $[F]$ gives the bracket in (103). Apply (98) at the input, and if desired at the output, to obtain both displays. □

The output projection is explicit: the theorem does not claim that classical substitution itself preserves $\beta = 0$. The scalar semigroup of Theorem 5.1, passed on a joint system as in Section 3, permits the finite choice $L = R^q$ after R, J are fixed. The bound is on a class admitting a newly justified localization, not on an arbitrarily fixed sequential carrier assignment.

7.6 Quantum descent is a separate obligation

No quantum map on $\mathcal{Q}_{a,b}$ has been defined here. Equality of infinite normalized totals on compact fields does not by itself identify finite-regulator exponentials, their sharp restricted integrals, chosen vacuum normalizations, or their limits. Thus this quotient does not yet permit arbitrary equivalent representatives to replace the actual quantum output inside its next finite integration. Quantum descent and its continuity and normalization compatibility, the full amplitude recursion, and the coupling trajectory remain separate.

8 The interface with a renormalized quantum induction

The low-degree classification is now tied to the actual local symmetries and a normalized classical reference. The exact coupling rebase is

$$g_{\mathrm{next}}^{-2} = g^{-2} - \beta, \qquad E_{\mathrm{ren}} = E - E(0) - \beta S_{MR}.$$

This is an identity of the density after extraction. A sign or trajectory estimate for the generated β is still needed; the identity does not supply one.

There is a concrete route to the spatial part of a contraction. Fix the elementary macrostep R and its sufficient auxiliary grouping J. The spatial remainder estimate has constants depending on these fixed data. Select a finite q afterward, so that its endpoint ratio $L = R^q$ makes $C(J)L^{-\alpha}$ small. The classical semigroup identifies the scalar inherited background with the direct endpoint background. Its new connected localization can then be estimated directly. The high-degree piece has the separate inverse-L estimate of Notebook 28, with its analytic radius chosen after this finite L.

The analytic compactness and quotient constructions make this classical change of representation precise. They supply joint limiting interactions and maps, a complete quotient, and a bounded

full-reference projection. The final classical estimate is a contraction once the finite endpoint ratio is sufficiently large and its common radius is chosen.

To turn this into a quantum induction, four interfaces must be established. First, the limiting interaction space must be matched to the regulators used by the actual quantum update, including winding terms and its vacuum normalization. Second, prove that the quantum update descends continuously to the quotient; classical descent alone does not identify the sharp finite-regulator integrals for equivalent inputs. Third, the q successive quantum steps must retain all intermediate generated terms, cutoffs and normalizations; the classical semigroup does not replace them by a single fluctuation integral. Fourth, the resulting recurrence must control both the renormalized activity and the actual coupling trajectory uniformly in the number of blocks of steps.

The present results supply the nonlinear marginal identity, the classical spatial gain and its representation space under stated hypotheses. The larger existence proof still needs the mixed and large-field histories, sourced observable estimates, continuum nontriviality, Euclidean symmetry and reconstruction already recorded in Notebook 01. The mass gap remains a subsequent spectral assertion.

References

[1] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 07, 15–16 and 25–28 in this workspace. The flat classical normalization, ordinary minimum theorem, strict classical support families and common-domain quantum macrostep are the principal prior inputs.

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions*, Commun. Math. Phys. **109** (1987), 249–301. Publisher record; Original article copy. The averaging symmetries and total Ward/subtraction calculations are checked against this primary source. Its stronger coefficientwise orbit hypotheses are not assumed here.

[3] T. Bałaban, *Renormalization group approach to lattice gauge field theories. II. Cluster expansions*, Commun. Math. Phys. **116** (1988), 1–22. Publisher record.

[4] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited raw accounts at commit `e392ef45013a581664bab76b90867170b30fe10e`.